

NBA JAM

Joystick & Button Restoration Guide

Modern Replica 3/4-Size LED Cabinet • Controls • Actuators • E-Clips • Reliability

Prepared for Coin Op 4 Charity / Arcade Angels

Practical notes so restored replica cabinets stay reliable in youth centers and community spaces

1. Overview & Goals

This guide covers joystick and push-button service on a **modern replica 3/4-size LED NBA Jam** cabinet — the kind commonly sold at department stores as a home / family arcade. The original consumer brand behind many of these cabinets is now defunct and restructuring. Official spare-parts support is therefore inconsistent. **Always check on manufacturer parts first** (current owner, successor company, authorized service, or remaining warehouse stock). If that channel has no support for the specific actuator, e-clip, spring, microswitch, or button plunger you need, rebuild with standard arcade hardware or print the missing plastic pieces.

For public and youth-center placements the priorities are reliability, safety, hygiene, and a positive, crisp click on every direction and shot button. NBA Jam is a physical game. Players mash Shoot, Pass, and Turbo. Stock replica controls were designed for living-room use, not all-day community play. Rebuild them before placement rather than after the first weekend of complaints.

Scope of this document

Focused service of the four-player control deck: joysticks (shaft, actuator, spring, e-clip, restrictor, microswitches) and action / start buttons. Cabinet cosmetics, LED monitor, and main board work are outside this guide except where they affect control access or testing.

2. Safety

- Unplug the cabinet and disconnect the power brick before lifting the control deck or pulling connectors.
- These replica decks use low-voltage control wiring, but the LED monitor and power supply still present electrical risk. Do not service with the unit powered.
- E-clips are spring-loaded. Wear eye protection. Work over a towel or parts tray so a flying clip is recoverable.
- Keep small parts (e-clips, washers, nuts, screws) away from younger helpers until they are bagged.
- Do not use excessive force on plastic restrictor plates or PCB-mounted switch boards. They crack.
- If a control harness uses a ribbon or latching connector, release the latch — do not yank the cable.

3. Accessing the Control Deck

On typical 3/4-size replica NBA Jam cabinets the entire control panel lifts after removing the screws along the underside or rear lip of the deck (commonly Phillips). Support the deck as the last screws come out so it does not drop on the harness. Photograph every connector before unplugging. Label Player 1–4 looms. Set the deck face-down on a padded surface with the bat-tops hanging off the edge, or rest it on two blocks so the sticks are not loaded sideways.

- Photograph the underside before any disassembly. Replica wiring is not always color-standardized.
- Note whether each stick uses discrete microswitches, a small PCB with four switches, or a 5-pin header.
- Bag hardware per player (P1, P2, P3, P4). Mixing worn actuators across players hides the bad one.
- If the overlay is peeling, tape the edges before flipping the deck so dirt does not get under the art.

4. Joystick Anatomy (Replica Midway-Style Stack)

Most department-store NBA Jam replicas use a compact American / Zippy-style stick rather than a full-size Happ Competition or Sanwa JLF. The stack from the player side down is typically:

Part	What it does	Failure mode in public use
Bat top / ball top	Player grip. Usually threaded onto the shaft (M6 or similar).	Backs off during dunk contests. Shaft slots, if present, let you tighten against a blade.
Shaft + pivot / grommet	The metal or plated shaft rides in a plastic or rubber pivot.	Oval wear = slop. Dry pivot = scratchy feel. Cracked grommet = off-center rest.
Restrictor plate	Limits throw to 4-way or 8-way. NBA Jam is a 4-direction game.	Cracked corners, missing screws, or an 8-way gate that makes diagonals too easy.
Actuator (cam / spring base)	Square or diamond plastic block that presses the four microswitches.	Corners round off. Then directions go dead, double-trigger, or feel mushy.
Return spring	Centers the shaft when released.	Takes a set, rusts, or collapses. Stick leans and holds a direction.
Washer + e-clip (e-ring)	Retains the entire spring / actuator stack on the shaft groove.	Clip flies off, bends, or was never fully seated. Stack drops; stick dies.
Microswitches (×4)	NO contacts closed by the actuator.	Worn levers, cracked bodies, cold solder on PCB versions, oxidized contacts.

The actuator is the part most people miss. It is the plastic piece under the restrictor that the spring sits against (sometimes called the spring base, actuator cam, or “white square”). When its corners wear, no amount of switch cleaning restores a clean 4-way.

5. Common Issues Seen on These Replica Decks

Joystick

- **Dead one direction.** Worn actuator corner, bent switch lever, unseated header, or a cracked copper trace on cheap switch PCBs.
- **Stick leans and the player keeps running.** Weak spring, missing e-clip, or actuator no longer square to the switches.
- **Huge deadband / slop.** Rounded actuator, worn pivot, loose mounting nuts, or a restrictor held by stripped screws.
- **Scratchy or gritty throw.** Dust, soda, and carpet grit packed around the pivot. Dry plastic-on-plastic contact.
- **Bat top spins off mid-game.** Factory thread locker failed. Tighten against shaft flats or slots; a drop of removable (blue) thread locker is enough.
- **E-clip missing after a prior “repair.”** The last person compressed the spring, the clip launched, and the stick was reassembled on hope.
- **Intermittent directions when the deck is flexed.** Ribbon or 5-pin connector not fully seated; strain relief missing.

Buttons (Shoot / Pass / Turbo / Start and any extra deck buttons)

- **Stuck down.** Sticky soda under the plunger, swollen cheap plastic, or a switch lever folded under the plunger.
- **No click, no register.** Dead microswitch, unplugged .110" QD terminal, or broken common wire in the daisy-chain.
- **Double-taps / ghost shots.** Switch lever bent too close to the plunger; actuator travel on the button is too long.
- **LED buttons dark (if equipped).** Separate LED leads reversed or unplugged. The switch can still work with a dead lamp.
- **Nut backing off under the deck.** Vibration plus a thin stamped nut. Use a proper nylon-insert nut or a drop of removable locker.

6. Tools & Consumables

- Phillips and small flat screwdrivers; needle-nose pliers; optional dedicated e-clip installer/remover (6–9 mm sizes).
- Eye protection, magnetic parts tray, painter’s tape for labels, phone camera.
- Electrical contact cleaner (plastic-safe), isopropyl alcohol, lint-free cloths. No oil on switch contacts.
- Light silicone or lithium grease — *pivot and shaft only*, never on microswitch points.

- Multimeter with continuity beep for switch and harness checks.
- Spare e-clips in the shaft size you measure (do not assume; replica shafts vary).
- Spare microswitches (standard arcade levered switches, ~28 × 16 mm body is common).
- Calipers if you will 3D-print an actuator. Measure shaft diameter, groove height, and actuator thickness.

7. Joystick Disassembly — E-Clip Procedure

The e-clip (e-ring) is the single most-lost part in this repair. It lives in a machined or molded groove at the bottom of the shaft and keeps the actuator and spring from falling off. The spring is under compression the entire time you are working. Plan the shot before you pick the clip.

1. **Remove the bat top** if you need the shaft out through the panel. Grip any flats or rear slots on the shaft so you do not twist the switches. A loose top can often be tightened without a full teardown.
2. **Unplug the stick harness** (5-pin header or four discrete QD terminals). Photograph orientation. On PCB sticks, one pin is usually common.
3. **Unbolt the mounting plate** from the underside of the deck. Support the stick so it does not drop.
4. **Set the assembly on the bench, actuator facing you.** Identify the e-clip, any flat washer, the actuator, and the spring.
5. **Compress the actuator and spring** toward the mounting plate with thumb and forefinger. This unloads the clip. If you skip this step the clip will launch.
6. **Remove the e-clip.** Preferred: matching e-clip tool hooked under the open side. Acceptable: small flat driver, work one ear off the groove, contain the clip with your other hand. Do this inside a clear bag if you have already lost one today.
7. **Slide off washer → actuator → spring.** Note which face of the actuator points at the switches. Many replica actuators are not symmetrical; a 180° flip changes engage distance.
8. **Inspect the shaft groove.** A mushroomed or burred groove will not hold a new clip. Dress lightly with a fine file. Do not deepen the groove.

E-clip field rules

Never reuse a spread or twisted clip. A distorted ring will walk off the groove the first time a player leans on Turbo. Keep a small bag of new clips in the machine's spare-parts envelope. If the original clip size is unknown, measure the groove diameter with calipers and match a standard external e-ring. Typical replica sticks use a small clip in the 6–9 mm family — verify, do not guess. On reassembly, you must hear or feel the clip snap fully into the groove all the way around, not just on one ear.

8. Actuator Replacement in Detail

The actuator is the square (sometimes slightly diamond or radiused-square) block that converts stick angle into four discrete switch presses. On replica NBA Jam decks this part is almost always a soft or medium-hard plastic. After a season of community play the corners radius off. Symptoms: a direction that needs extra throw, two directions that fire together, or a stick that “feels fine” but drops inputs in a fast break.

Inspect before you print or order

- Set the worn actuator on a flat card and look at the silhouette. If the corners are visibly round, replace it.
- Compare thickness to a known-good player's actuator on the same cabinet. Thickness sets pre-load on the switches.
- Check the bore. An egg-shaped hole on the shaft means the actuator is spinning and will never stay timed to the switches.
- Check the spring pocket (if the actuator doubles as the spring base). A cracked pocket lets the spring walk sideways.

Replacement order of operations

- **Check on manufacturer parts.** Ask the current rights-holder / successor / remaining parts desk for the joystick service kit or the individual actuator. Quote the cabinet model printed on the back of the deck or the rating label. Because the original consumer brand is defunct and restructuring, expect slow replies or a “no longer stocked” answer.
- If a factory actuator is available, use it. Fit is already correct for that shaft and restrictor.

- If the factory stick is a known Zippy-pattern or other common compact stick, a commercial aftermarket actuator / spring-base for that pattern is the next choice.
- If neither channel has the part, measure and 3D-print (Section 10) or upgrade the entire stick to a serviceable full-size 4-way with a plate adapter — only if the deck depth and overlay holes allow it.

Fitment notes that decide whether the repair works

- Bore must be a close slip on the shaft. Too loose = actuator rotates and hits two switches. Too tight = you will never get the e-clip on.
- Corner profile should be sharp enough to hit one switch cleanly at the end of a 4-way throw, with a small amount of over-travel. A printed part that is oversized will stack-switch (Up+Left at once).
- Height / thickness must leave enough groove showing to accept the washer and e-clip after the spring is compressed. If a printed actuator is even 0.5 mm too tall, the clip will not seat. File the top face, do not grind the shaft.
- Some replica actuators include a small step that registers inside the spring. Reproduce that step or the spring will cock and the stick will lean.
- Orient flats to the four switches, not to the diagonal. NBA Jam wants cardinal directions.

9. Springs, Pivots, Restrictors & Microswitches

Spring. If the stick does not snap back to center, replace the spring. Stretching a tired spring is a short-term field trick only — it relaxes again. Note the free length and wire diameter before you toss the original. A slightly stronger spring is acceptable for public use; an overly heavy spring makes younger players fight the stick and hammers the actuator.

Pivot / grommet. Clean packed dust with isopropyl. A dry, squeaking pivot gets a *thin* film of silicone grease. Do not pack grease into the switch area. Cracked grommets should be replaced or printed in TPU rather than rigid PLA.

Restrictor. NBA Jam is four directions. If the replica shipped with a sloppy 8-way gate, a 4-way restrictor (factory, aftermarket, or printed) reduces accidental diagonals and is kinder to the actuator corners. Cracked restrictors must be replaced — a missing corner is how sticks develop a “favorite” direction.

Microswitches. Test each switch off the harness with a continuity beep while you press the lever. A good switch has a sharp click and a clean make/break. Replace any switch that is mushy, intermittent, or has a cracked hinge. On PCB-mounted replica sticks, inspect the solder side for hairline cracks at the common pin — a very common failure after the deck has been flexed. Reflow or run a short jumper rather than condemning the whole stick if the traces are otherwise intact.

10. 3D-Printing Parts When Manufacturer Support Is Gone

Use printing as a support strategy, not a first resort. Check on manufacturer parts. If the successor company, a remaining warehouse, or a documented compatible commercial part can supply the actuator, clip, spring, or plunger, buy that. Print when the alternative is a dead player station.

Parts worth printing on this cabinet

- Joystick actuator / spring base (highest value part).
- 4-way restrictor plate sized to the existing bolt pattern.
- Pivot bushing or dust washer (TPU for the bushing, PETG for washers).
- Button plunger or the thin plastic collar that centers a cheap plunger, if the original is cracked.
- Underside reinforcement plate or spacer if the particle-board deck is crushed around a stick hole.
- E-clip is **not** a good printed part. Use real spring steel.

How to model the actuator if no file exists

- Caliper the shaft diameter (often in the 8–10 mm range on compact replica sticks — measure yours).
- Measure actuator outside size across flats, thickness, spring-pocket diameter and depth, and the distance from the shaft axis to each switch lever at rest.
- Photograph the part on graph paper from square-on. Import as a canvas and trace.

- Add a small chamfer on the bore so it starts on the shaft easily. Keep the switch-facing corners crisp.
- Print a test washer first to confirm bore fit before committing a full actuator.

Print settings that survive youth-center use

- **Material:** PETG or nylon for actuators and restrictors. PLA is acceptable for a same-day emergency and should be scheduled for replacement. TPU 95A for grommets / bushes only.
- **Walls / infill:** 4–6 walls, 30–50% infill on actuators. These parts take impact on every dunk.
- **Orientation:** Bore vertical so layer lines do not split across the corners that hit the switches.
- **Tolerance:** First print slightly thick. File the top face until the e-clip groove is fully visible with the spring compressed. Filing is faster than reprinting a short part.
- **Post-process:** Knock off stray filaments on the bore. Do not round the working corners with sandpaper “to make it pretty.”

Reliability over originality

Coin Op 4 Charity placements favor parts that a volunteer can service again in six months. A printed actuator that is documented (file name, material, date, shaft diameter) in the host-site envelope is better than a mystery factory piece with no spare. Keep the worn original in the bag as a pattern.

11. Button Repair

NBA Jam uses three action buttons per player plus Start (Shoot, Pass, Turbo on the original layout). Replica decks usually use snap-in or nutted 24–30 mm buttons with a separate microswitch clipped to the housing. Treat buttons as a hygiene item as much as a mechanical one — sticky plungers are how machines feel “dirty” even when the cabinet is clean.

Removal

- Unplug both .110" terminals (and LED leads if present). Photograph which terminal is common / NO. Swapping them is harmless on a simple NO switch but will confuse you during testing.
- Nutted buttons: hold the plunger, spin off the ring nut. Snap-in buttons: pinch the two locking ears and push out through the overlay.
- If the overlay hole is torn, do not yank. Cut the nut free and plan a backing washer on reinstall.

Service

- Pull the plunger and the return spring. Wash plunger and housing with mild soap; rinse; dry completely. Soda residue is the number-one sticky-button cause.
- Inspect the plunger shaft for flat spots or a melted lip. Replace or print if it binds in the housing.
- Click-test the microswitch off the button. Replace if the click is dull or continuity is dirty. Do not spray oil into the switch body.
- If the switch lever is bent downward, the button will feel “already pressed.” Bend only as a last resort; replacement is cleaner.
- On LED buttons, confirm polarity when reconnecting. A dark lamp with a working click is an LED/wiring issue, not a game-board issue.
- Check on manufacturer parts for complete button assemblies first. If the successor has no stock, standard 24 mm or 30 mm arcade buttons with screw terminals are a durable substitute provided the hole size matches. Use a printed or commercial adapter collar if the replica hole is an odd size.

Reinstall

- Confirm the plunger returns on its own before the deck goes back on the cabinet.
- Do not overtighten ring nuts into particle board — you will crush the deck and oval the hole.
- Route wires so the deck hinge or last mounting screw cannot pinch a Turbo lead. A pinched Turbo wire presents as “random dunks.”

- If a nut will not stay put, a printed underside washer plus a nyloc-style nut ends the callback.

12. Reassembly Sequence (Stick)

1. Pivot and restrictor on the mounting plate; screws snug, not stripped.
2. Shaft through the panel / plate from the player side.
3. Spring onto the shaft, correct end into the actuator pocket.
4. Actuator on, flats aimed at the four switches, pocket facing the spring.
5. Flat washer (if used) on top of the actuator.
6. Compress spring + actuator until the shaft groove is fully exposed. This is a two-hand job. A helper is worth it.
7. Start the e-clip onto the groove with the open side last. Press until it snaps around the full diameter. Tug the actuator — it must not be able to pass the clip.
8. Confirm the shaft returns to center and each direction clicks exactly one switch.
9. Reconnect the harness. Strain-relieve with a zip-tie to the plate, not to a switch body.
10. Bolt the plate back to the deck. Check that the bat top still clears the overlay and that the stick is not cocked because one mounting screw is longer than the others.

13. Electrical Quick Checks

- With the cabinet still unplugged, beep each switch at the player-side connector: open at rest, closed when held.
- Wiggle the harness while beeping. Intermittents here are connectors, not actuators.
- If one entire player is dead, look at that player's header on the deck PCB or encoder before rebuilding a good stick.
- Do not reflow or "touch up" the main game board as a first step for a single dead direction. Prove the control first.

14. Final Testing Checklist Before Placement

- Cold boot to attract mode. All four player stations light / respond as designed.
- Each joystick: clean Up, Down, Left, Right. No held direction at rest. No diagonal when the player intends a cardinal move.
- Each Shoot, Pass, Turbo, and Start on all seated players registers every time, including rapid Turbo taps.
- No bat top rotates off during a full game. No button nut audible as it spins.
- Play at least one full game on each stick (or a long practice / dunk sequence) — not just a menu wiggle.
- Open and close the deck once after the test. Confirm no harness pinch and that every e-clip is still seated.
- Cabinet exterior clean, level, no sharp screw points under the deck lip.
- Free-play or site pricing configured and written down.
- Spare e-clips, one spare actuator (printed or factory), two spare microswitches, and a spare button plunger bagged and labeled for the host.

15. Notes for Host Sites (Youth Centers & Community Spaces)

- Provide a one-page sheet: power on/off, free-play status, and who to call. Do not ask staff to open the deck.
- Weekly: wipe the control panel and bat tops. Sticky soda on Turbo is the start of a stuck-button call.
- Ask staff to report a leaning stick or a silent button the same day. An e-clip that has started to walk off is a five-minute fix if caught early and a lost part if it hits the floor.
- Document every non-original part: "P2 actuator PETG printed 2026-09, shaft 9 mm, file actuator_nbajam_v2.stl" so the next volunteer is not guessing.
- If the manufacturer / successor later restocks a genuine kit, prefer that part on the next service pass and keep the printed piece as a travel spare.

16. Parts Policy — Short Version

- 1) Check on manufacturer parts (including any company that now holds the catalog while the original consumer brand restructures).
- 2) If the answer is no stock and no timeline, use standard arcade microswitches, springs, e-clips, and buttons that a volunteer can buy locally.
- 3) Print actuators, restrictors, bushings, and odd-size collars only when those two doors are closed.
- 4) Never print the e-clip.
- 5) Write down what you did.

This guide is intended for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. The goal is a replica NBA Jam control deck that stays reliable, safe, and fun for the people who actually play it — not a museum-original living-room toy.

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